Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

The cubic polynomial $g(x) = x^3 + ax^2 + bx - 6$ has a factor $(x+1)$. When $g(x)$ is divided by $(x-2)$ the remainder is 12.

(a) Find the values of a and b. **[5]**

(b) Express $g(x)$ as the product of a linear factor and a quadratic factor. **[2]**

(c) Show that the equation $g(x) = 0$ has three distinct real roots. **[2]**

2 The curve C has equation $y = f(x)$ where $f(x) = (x-1)(x+2)(x-3)$.

(a) Write down the coordinates of the points where C meets the x-axis. **[1]**

(b) The curve is translated by $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$. Find the equation of the new curve in fully factorised form. **[3]**

(c) The curve $y = f\left(\frac{1}{2}x\right)$ is drawn. Find the coordinates of the points where it meets the x-axis. **[3]**

Question 1

(a) $g(-1) = 0$: $-1 + a - b - 6 = 0 \Rightarrow a - b = 7$. $g(2) = 12$: $8 + 4a + 2b - 6 = 12 \Rightarrow 2a + b = 5$. Adding: $3a = 12$, so $a = 4$ and $b = -3$

(b) $g(x) = x^3 + 4x^2 - 3x - 6 = (x+1)(x^2 + 3x - 6)$

(c) For $x^2 + 3x - 6 = 0$: $b^2 - 4ac = 9 + 24 = 33 > 0$, so two distinct real roots. Neither equals -1 (since $1 - 3 - 6 \neq 0$), so there are three distinct real roots in total

Question 2

(a) $(1,0)$, $(-2,0)$ and $(3,0)$

(b) Translation by $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ gives $y = f(x+2) = (x+1)(x+4)(x-1)$

(c) $f(\frac{1}{2}x) = 0 \Rightarrow \frac{1}{2}x = 1, -2, 3 \Rightarrow x = 2, -4, 6$. The points are $(2,0)$, $(-4,0)$ and $(6,0)$